Data Mining

Jobsheet Association Rule

Muhammad Annora Ega Prasetyo

244107060069

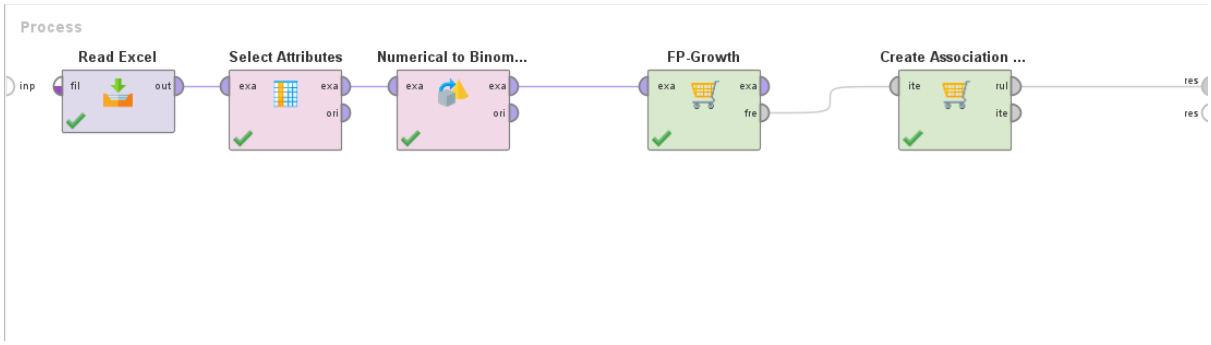
SIB 2C

Tugas

1. Gunakan dataset pada slide (PPT). Modifikasi format dataset menjadi seperti Grocery.xlsx.

TID	1	2	3	4	5
100	1	0	1	1	0
200	0	1	1	0	1
300	1	1	1	0	1
400	0	1	0	0	1
500	1	0	1	0	1

2. Lakukan association rule mining menggunakan Altair AI Studio lalu bandingkan hasilnya dengan hasil pada slide (screenshot design serta itemset dan rule yang dihasilkan).



No.	Premises	Conclusion	Support	Confidence	LaPlace	Gain	p-s	Lift	Co
9	3.0	5.0	0.600	0.750	0.889	-1	-0.040	0.938	0.
10	5.0	3.0	0.600	0.750	0.889	-1	-0.040	0.938	0.
11	3.0	1.0	0.600	0.750	0.889	-1	0.120	1.250	1.
12	5.0	2.0	0.600	0.750	0.889	-1	0.120	1.250	1.
13	1.0	3.0	0.600	1	1	-0.600	0.120	1.250	∞
14	4.0	3.0	0.200	1	1	-0.200	0.040	1.250	∞
15	2.0	5.0	0.600	1	1	-0.600	0.120	1.250	∞
16	4.0	1.0	0.200	1	1	-0.200	0.080	1.667	∞
17	5.0, 1.0	3.0	0.400	1	1	-0.400	0.080	1.250	∞
18	3.0, 2.0	5.0	0.400	1	1	-0.400	0.080	1.250	∞
19	1.0, 2.0	3.0 5.0	0.200	1	1	-0.200	0.040	1.250	∞
20	4.0	3.0, 1.0	0.200	1	1	-0.200	0.080	1.667	∞
21	3.0, 4.0	1.0	0.200	1	1	-0.200	0.080	1.667	∞
22	1.0, 4.0	3.0	0.200	1	1	-0.200	0.040	1.250	∞
23	1.0, 2.0	5.0	0.200	1	1	-0.200	0.040	1.250	∞
24	1.0, 2.0	3.0, 5.0	0.200	1	1	-0.200	0.080	1.667	∞
25	3.0, 1.0, 2.0	5.0	0.200	1	1	-0.200	0.040	1.250	∞
26	5.0, 1.0, 2.0	3.0	0.200	1	1	-0.200	0.040	1.250	∞

3. Lakukan association rule mining menggunakan Python lalu bandingkan hasilnya. Screenshot code dan rule yang dihasilkan.

```
============================================================
DATASET (One-Hot Encoding)
============================================================
       1      2      3      4      5
0   True  False   True   True  False
1  False   True   True  False   True
2   True   True   True  False   True
3  False   True  False  False   True
4   True  False   True  False   True

============================================================
FREQUENT ITEMSETS
============================================================
```

```
============================================================
FREQUENT ITEMSETS
============================================================

--- Apriori (slide dosen) ---
    support              itemsets  support_count
0       0.8        frozenset({3})              4
1       0.8        frozenset({5})              4
2       0.6        frozenset({1})              3
3       0.6        frozenset({2})              3
4       0.6     frozenset({1, 3})              3
5       0.6     frozenset({5, 3})              3
6       0.6     frozenset({5, 2})              3
7       0.4     frozenset({2, 3})              2
8       0.4     frozenset({1, 5})              2
9       0.4  frozenset({1, 5, 3})              2
10      0.4  frozenset({5, 2, 3})              2
```

```
--- FP-Growth (Altair AI Studio) ---
    support              itemsets  support_count
0       0.8        frozenset({3})              4
1       0.8        frozenset({5})              4
2       0.6        frozenset({1})              3
3       0.6        frozenset({2})              3
4       0.6     frozenset({5, 3})              3
5       0.6     frozenset({1, 3})              3
6       0.6     frozenset({5, 2})              3
7       0.4     frozenset({1, 5})              2
8       0.4  frozenset({1, 5, 3})              2
9       0.4     frozenset({2, 3})              2
10      0.4  frozenset({5, 2, 3})              2
```

```
=====================================================
ASSOCIATION RULES
=====================================================

--- Apriori (slide dosen) ---
          antecedents        consequents  support  confidence      lift
0       frozenset({1})     frozenset({3})      0.6    1.000000  1.250000
1       frozenset({2})     frozenset({5})      0.6    1.000000  1.250000
2    frozenset({2, 3})     frozenset({5})      0.4    1.000000  1.250000
3    frozenset({1, 5})     frozenset({3})      0.4    1.000000  1.250000
4       frozenset({3})     frozenset({1})      0.6    0.750000  1.250000
5       frozenset({5})     frozenset({3})      0.6    0.750000  0.937500
6       frozenset({5})     frozenset({2})      0.6    0.750000  1.250000
7       frozenset({3})     frozenset({5})      0.6    0.750000  0.937500
8    frozenset({1, 3})     frozenset({5})      0.4    0.666667  0.833333
9       frozenset({2})     frozenset({3})      0.4    0.666667  0.833333
10   frozenset({5, 3})     frozenset({1})      0.4    0.666667  1.111111
11      frozenset({1})  frozenset({5, 3})      0.4    0.666667  1.111111
12   frozenset({5, 2})     frozenset({3})      0.4    0.666667  0.833333
13      frozenset({1})     frozenset({5})      0.4    0.666667  0.833333
14      frozenset({2})  frozenset({5, 3})      0.4    0.666667  1.111111
```

```
15  frozenset({5, 3})     frozenset({2})      0.4    0.666667  1.111111
16     frozenset({3})     frozenset({2})      0.4    0.500000  0.833333
17     frozenset({5})     frozenset({1})      0.4    0.500000  0.833333
18     frozenset({5})  frozenset({1, 3})      0.4    0.500000  0.833333
19     frozenset({3})  frozenset({1, 5})      0.4    0.500000  1.250000
20     frozenset({5})  frozenset({2, 3})      0.4    0.500000  1.250000
21     frozenset({3})  frozenset({5, 2})      0.4    0.500000  0.833333
```

```
--- FP-Growth (Altair AI Studio) ---
          antecedents        consequents  support  confidence      lift
0      frozenset({2})     frozenset({5})      0.6    1.000000  1.250000
1      frozenset({1})     frozenset({3})      0.6    1.000000  1.250000
2   frozenset({2, 3})     frozenset({5})      0.4    1.000000  1.250000
3   frozenset({1, 5})     frozenset({3})      0.4    1.000000  1.250000
4      frozenset({3})     frozenset({5})      0.6    0.750000  0.937500
5      frozenset({5})     frozenset({3})      0.6    0.750000  0.937500
6      frozenset({5})     frozenset({2})      0.6    0.750000  1.250000
7      frozenset({3})     frozenset({1})      0.6    0.750000  1.250000
8   frozenset({1, 3})     frozenset({5})      0.4    0.666667  0.833333
9      frozenset({1})     frozenset({5})      0.4    0.666667  0.833333
10     frozenset({1})  frozenset({5, 3})      0.4    0.666667  1.111111
11  frozenset({5, 2})     frozenset({3})      0.4    0.666667  0.833333
12     frozenset({2})     frozenset({3})      0.4    0.666667  0.833333
13  frozenset({5, 3})     frozenset({1})      0.4    0.666667  1.111111
14     frozenset({2})  frozenset({5, 3})      0.4    0.666667  1.111111
15  frozenset({5, 3})     frozenset({2})      0.4    0.666667  1.111111
16     frozenset({5})     frozenset({1})      0.4    0.500000  0.833333
17     frozenset({5})  frozenset({1, 3})      0.4    0.500000  0.833333
18     frozenset({3})  frozenset({1, 5})      0.4    0.500000  1.250000
19     frozenset({3})     frozenset({2})      0.4    0.500000  0.833333
20     frozenset({5})  frozenset({2, 3})      0.4    0.500000  1.250000
```

```
21     frozenset({3})  frozenset({5, 2})      0.4    0.500000  0.833333

==========================================================
PERBANDINGAN
==========================================================
Frequent itemset SAMA antara Apriori dan FP-Growth
Jumlah rules Apriori   : 22
Jumlah rules FP-Growth : 22
```